ECON 8803: PS3

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1 Empirical Problems

1. IV four ways: Consider the just-identified case with $J = K = 1$. Using any dataset of your choice and with any statistical software, verify that estimating the IV using the following four different ways yields numerically equivalent results (do not worry about the violation of exclusion restriction).

   - Dataset: Angrist and Krueger 1991

	Specification	**IV Estimate**
a	Regular IV	0.102
b	2SLS	0.102
c	Ratio	0.102
d	Moment Condition	0.102

2. Many (Weak) IV: For an observed random sample of $n = 250$ observations $(y_i, x_i, z_i') \in \mathbb{R}^{1+1+l}$, consider:

$$y_i = x_i\beta_0 + \varepsilon_i$$
$$x_i = z_i'\pi_0 + u_i \quad (i = 1, ..., n)$$

where, $l = 15, \beta_0 = 0, \pi_0 = \frac{(1,...,1)'}{\sqrt{n}}$ and

$$(\varepsilon_i, u_i)' \sim \mathcal{N}\left(0, \begin{bmatrix} 1 & -0.7 \\ 0.7 & 1 \end{bmatrix}\right)$$
$$z_i \sim \mathcal{N}[0, I_l], \quad (I \text{ is identity matrix})$$

   (a) In large samples with multiple IVs, the first-stage F-statistic is close to:

$$1 + \frac{n\mathbb{E}[(z_i'\pi_0)^2]}{l\sigma_u^2}$$

Calculate this value.

$$\begin{aligned}
F &= 1 + \frac{n\mathbb{E}[(z_i'\pi_0)^2]}{l\sigma_u^2} \\
&= 1 + \frac{n\pi_0'\mathbb{E}[z_i'z_i]\pi_0}{l\sigma_u^2} \\
&= 1 + \frac{n\pi_0'Var(z_i) + \mathbb{E}[z_i]\pi_0}{l\sigma_u^2} \\
&= 1 + \frac{n\pi_0'\mathbf{I}_l\pi_0}{l\sigma_u^2} \\
&= 1 + \frac{n\pi_0^2}{l\sigma_u^2} \\
&= 1 + \frac{n(\frac{l}{n})}{l\sigma_u^2} \\
&= 1 + \frac{l}{l} = 1 + 1 = \boxed{2}
\end{aligned}$$

(b) In a statistical software of your choice, generate data from the above model and calculate the following statistics: $\hat{\beta}_{OLS}, \hat{\beta}_{2SLS}, \hat{F}, \hat{\beta}_{JIVE}$. *Hint*: $\hat{\beta}_{JIVE}$ is the jackknife IV estimator (JIVE), which avoids overfitting of the first-stage by a leave-one-out procedure.

- For each i, estimate the first-stage $\hat{\pi}_{-1}$ excluding observation i.
- Obtain fitted values $\tilde{D}_i = z_i'\hat{\pi}_{-1}$ and set $\hat{\beta}_{JIVE} = (\tilde{D}'D)^{-1}\tilde{D}Y$ (note that in this question, x_i is the endogenous variable)
- Instead of running the first stage n times, you may calculate $\tilde{D}_i = z_i'\hat{\pi}_{-i} = \frac{z_i'\hat{\pi} - P_{ii}x_i}{1-P_{ii}}$ where $P_{ii} = z_i'(\sum_{i=1}^n z_i z_i')^{-1}z_i$.
- Repeat (b) 10,000 times and report the simulated median of the coefficient estimates and the simulated mean of the F statistic. What do you observe?
  - As shown in the table below, both OLS and 2SLS are heavily biased estimators of the true estimate, while the jackknife estimator has almost literally no bias. By excluding the error term for each observation, jackknife reduces bias here when compared to 2SLS, which contains both e_{-i} and e_i.

Specification	**Estimates**
$\hat{\beta}_{OLS}$	0.660
$\hat{\beta}_{2SLS}$	0.344
$\hat{F}$	2.010
$\hat{\beta}_{JIVE}$	0.006
Control Function	0.344

(c) In the above simulation, verify that controlling the residuals from first stage as the "control function" yields the same estimate as 2SLS (i.e., calculate the median control function estimate).

- See above table.

3. Measurement error: For a random sample of $n = 250$ observations, consider the following model:

$$y_i = D_i\beta_0 + \varepsilon_i$$

where $\varepsilon_i \sim \mathcal{N}(0, 1)$. You may choose whatever distribution for D_i.

- Consider $\mathcal{D}_{1i} = D_i + \xi_i$, where $\xi_i \sim \mathcal{N}(0, \sigma_\xi^2)$
- Consider $\mathcal{D}_{2i} = D_i + \eta_i$, where $\eta_i \sim \mathcal{N}(0, \sigma_\eta^2)$

You may choose whatever parameter values for β_0, σ_ξ^2, and σ_η^2.

(a) Regress Y_i on D_i and report the coefficient estimate.

(b) Now, regress Y_i on D_{1i} and report the estimate. Show that this estimate is close to:

$$\beta_0 \frac{Var[D_i]}{Var[D_i] + \sigma_\xi^2}$$

- We set $\beta_0 = 2, \sigma = 1$. We would expect the estimate of regressing Y_i on D_{1i} to be $2\frac{1}{1+1} = 1$.

(c) Now obtain the coefficient estimate from instrumenting $\mathcal{D}_{1i}$ with $\mathcal{D}_{2i}$.

Using a statistical software of your choice, repeat the above process 10,000 times and report the median of the coefficient estimates. What do you observe? Write a short note relating your findings to the model and theory presented in class.

	Specification	**Median**
(a)	$Y_i \sim D_i$	2.001
(b)	$Y_i \sim \mathcal{D}_{1i}$	1.001
(c)	$\mathcal{D}_{1i} \sim \mathcal{D}_{2i}$	2.000

Given that we set β_0 to 2, it follows that the simulation pulls out $Y_i \sim D_i = 2$. Regressing Y_i on $\mathcal{D}_{1i}$ results in biased measurement error because $\mathcal{D}_i$ is correlated with $\varepsilon_i - \tau\xi_i$. Using $\mathcal{D}_{2i}$ as an instrument for $\mathcal{D}_{1i}$ recovers the true estimation since it is uncorrelated with the composite error term.